

The prescribed floating-potential boundary condition in `model600` Normalisation, sign convention, and the linear trace constraint

Derived from the JOREK source; every step checkable against a named file and line

Abstract

We impose $V_p - V_{\text{wall}} = \Lambda k_B T_e / e$ with $\Lambda = 3$ on the plasma potential at a material wall — the local zero-current (floating) limit of the sheath characteristic. This note derives, from the JOREK variables and the code’s own coordinate convention, the exact linear relation between the stream function u and the electron temperature that this condition becomes, and the two coefficients that enter the assembled matrix row. The condition is *linear* in the evolved variables, so it needs no Newton iteration, no clipping, no incidence-angle floor and no relaxation parameter. The one genuinely delicate step is the sign, which two comments in the current source state in opposite ways; we settle it from the coordinate basis rather than from either comment.

1 What is being imposed, and what is not

For a Maxwellian plasma at a surface drawing zero net current, the sheath drops

$$V_p - V_{\text{wall}} = \Lambda \frac{k_B T_e}{e}, \quad \Lambda = \ln \sqrt{\frac{m_i}{2\pi m_e}} \approx 3. \quad (1)$$

This is the *floating* potential: the value at which the retarded electron flux exactly cancels the ion saturation flux. Prescribing it is a deliberate reduction of the full Langmuir characteristic $j = j_{\text{sat}}(1 - e^{-X})$ to its single zero-current point.

Honest scope. Eq. (1) imposes a *local, zero-current* closure. It cannot carry a net wall current, so it cannot produce thermoelectric currents between targets, and it is not a model of a conducting vessel closing current through the wall. What it does give — and what the full characteristic also gives — is a wall potential that follows T_e , hence a tangential electric field and an $\mathbf{E} \times \mathbf{B}$ drift wherever T_e varies along the wall.

2 JOREK variables and the two normalisations

The reduced-MHD ansatz uses the poloidal flux ψ and the velocity stream function u . Two conversions to SI are needed. With $n_0 = \text{central_density} \times 10^{20}$, $m_i = \text{central_mass} \times m_u$ and $\rho_0 = n_0 m_i$,

$$V_p [\text{V}] = \frac{F_0 u}{\sqrt{\mu_0 \rho_0}}, \quad k_B T_e [\text{J}] = \frac{T_e^{\text{JOREK}}}{\mu_0 n_0}. \quad (2)$$

The second is JOREK’s pressure normalisation ($p = \rho T$ with ρ in units of ρ_0 and pressure in units of $1/\mu_0$). The first is the statement $\Phi = F_0 u$ in JOREK units, converted to volts by the velocity normalisation $1/\sqrt{\mu_0 \rho_0}$; its *sign* is the subject of the next section.

3 The sign, settled from the basis and not from a comment

3.1 Why this needs care

Two comments in the present source state opposite conventions: `mod_sheath.bc.f90:207` says $\Phi = -F_0 u$, while `mod_boundary_conditions.f90:876` says $\Phi = +F_0 u$. They cannot both be right, and the choice flips the direction of every $\mathbf{E} \times \mathbf{B}$ drift the boundary condition drives.

The ambiguity is genuine and not resolvable from the equations alone, because ψ and u each carry their own sign convention which absorbs the basis handedness. Writing the $\mathbf{E} \times \mathbf{B}$ velocity as

$$\mathbf{v}_E = \frac{\mathbf{E} \times \mathbf{B}}{B^2}, \quad \mathbf{E} = -\nabla\Phi, \quad \mathbf{B} \simeq \frac{F_0}{R} \mathbf{e}_\phi \implies \mathbf{v}_{E,\text{pol}} = -\frac{R}{F_0} (\nabla\Phi \times \mathbf{e}_\phi) \quad (3)$$

and comparing with the implemented $\mathbf{v}_{\text{pol}} = cR(\nabla u \times \mathbf{e}_\phi)$ gives $\Phi = -cF_0 u$. Everything therefore rests on the sign c , which depends on how $\nabla u \times \mathbf{e}_\phi$ is expanded — i.e. on the handedness of the basis.

3.2 The basis, from three independent places in the code

JOREK uses $(\mathbf{e}_R, \mathbf{e}_Z, \mathbf{e}_\phi)$ *right-handed*, so $\mathbf{e}_R \times \mathbf{e}_Z = +\mathbf{e}_\phi$ and ϕ runs clockwise viewed from above. This is not a comment; it is visible three ways:

1. `diagnostics/jorek2_povray.f90:102--103` and `vacuum/mod_vacuum_fields.f90:1172--1173` map $x = R \cos \phi$, $y = -R \sin \phi$. The minus fixes the handedness.
2. `particles/mod_fields.f90:590--596`: `rot_tmp` is the right-handed curl for (R, Z, ϕ) with scale factors $(1, 1, R)$. Applied to the code's own $\mathbf{A}$ it returns the code's own $\mathbf{B}$ exactly; in the standard (R, ϕ, Z) basis it returns $-\mathbf{B}$ in all three components, so the test discriminates.
3. `diagnostics/new_diag/mod_expression.f90:1354`: $J_{\text{tor}} = -z_j/R$, the third curl component in that basis.

3.3 The consequence

In this basis $\mathbf{e}_R \times \mathbf{e}_\phi = -\mathbf{e}_Z$ and $\mathbf{e}_Z \times \mathbf{e}_\phi = +\mathbf{e}_R$, so

$$\nabla u \times \mathbf{e}_\phi = (u_Z, -u_R). \quad (4)$$

The implemented poloidal velocity is, from `particles/mod_fields.f90:581--582`,

$$\mathbf{v}_{\text{pol}} = (-R u_Z, +R u_R) = -R (\nabla u \times \mathbf{e}_\phi), \quad (5)$$

i.e. $c = -1$, which is exactly the ansatz of Hölzl *et al.* 2021 (eq. 26). Hence

$$\boxed{\Phi = +F_0 u} \quad (6)$$

and the code's u equals the paper's. The $\Phi = -F_0 u$ comment arose from comparing the implemented components against the paper while implicitly using the *standard* (R, ϕ, Z) basis; the same components read as $-R \nabla u \times \mathbf{e}_\phi$ in JOREK's basis and $+R \nabla u \times \mathbf{e}_\phi$ in the standard one, and the discrepancy was attributed to u rather than to the basis.

Consequence for this implementation. `sheath_norm` in `mod_sheath.bc.f90` returns $a_n < 0$ and must *not* be reused. The floating-potential module carries its own normalisation helper with $a_n > 0$, and a unit test pins it.

4 From the physical condition to the JOREK trace constraint

Insert (2) into (1):

$$\frac{F_0 u}{\sqrt{\mu_0 \rho_0}} - V_{\text{wall}} = \frac{\Lambda}{e} \frac{T_e^{\text{JOREK}}}{\mu_0 n_0}. \quad (7)$$

Solving for u and introducing the two standard groups

$$a_n = \frac{2eF_0\sqrt{\mu_0\rho_0}}{m_i}, \quad v_w = e V_{\text{wall}} \mu_0 n_0, \quad (8)$$

gives, using $\rho_0 = n_0 m_i$,

$$u = C_T T_e + C_V V_{\text{wall}}, \quad C_T = \frac{2\Lambda}{a_n}, \quad C_V = \frac{2e\mu_0 n_0}{a_n} = \frac{\sqrt{\mu_0 \rho_0}}{F_0}. \quad (9)$$

Equivalently $u = 2(\Lambda T_e + v_w)/a_n$. Note a_n carries the sign of F_0 , so reversing the toroidal field reverses u while leaving the reconstructed physical potential unchanged — as it must, since (1) contains no field direction.

Single-temperature builds. When `with.TiTe` is false the model evolves $T = T_i + T_e$ with $T_e = T/2$, so C_T is halved: $u = \Lambda T/a_n + C_V V_{\text{wall}}$.

5 Why no linearisation is required

Eq. (9) is *affine* in the evolved variables (u, T_e) at fixed Λ . Writing the implicit step for increments $\delta u, \delta T_e$ about the current state,

$$\delta u - C_T \delta T_e = -(u^n - C_T T_e^n - C_V V_{\text{wall}}) \equiv -g^n, \quad (10)$$

which is exact: there is no neglected higher-order term, the Jacobian entries $(1, -C_T)$ are constants, and the residual g^n vanishes identically once the constraint is satisfied. This is the entire content of the boundary row.

This is the structural reason the condition is robust. The full characteristic $j = j_{\text{sat}}(1 - e^{-X})$ is nonlinear in u through $\exp$, saturates at j_{sat} , and has a Jacobian $\partial j / \partial u \propto j_{\text{sat}} e^{-X}$ that vanishes both in ion saturation *and* wherever $j_{\text{sat}} \rightarrow 0$ (grazing incidence). Every gate, slope, floor and trust region in the current-carrying implementation exists to manage that. Eq. (9) has none of those failure modes: its Jacobian is the constant $(1, -C_T)$, it never inverts a characteristic, and it never divides by B_n , $g(b_n)$, j_{sat} or the element Jacobian.

6 Which degrees of freedom the constraint acts on

Because Λ is a constant, (9) may be differentiated along the boundary degree of freedom by degree of freedom:

$$u^{\text{val}} - C_T T_e^{\text{val}} = C_V V_{\text{wall}}, \quad u^{\partial_\ell} - C_T T_e^{\partial_\ell} = 0, \quad u^{\partial_\ell^2} - C_T T_e^{\partial_\ell^2} = 0, \dots \quad (11)$$

where ∂_ℓ denotes the pure tangential derivatives carried by the trace space. The constant V_{wall} appears only in the value equation, and only in the axisymmetric harmonic.

Two properties follow, and they are the reason this closure is immune to the grid pathologies that defeated the current-carrying route:

- **No geometry enters.** The constraint never forms B_n , never divides by the element Jacobian x_{jac} , and never converts logical derivatives into (R, Z) . The u and T_e trace DOFs are stored in the *same* nodal frame, so whatever that frame is — including a near-degenerate one — its scaling cancels identically between the two terms.
- **No normal derivative is touched.** Only the value and pure tangential DOFs are constrained; the normal and mixed DOFs remain governed by the PDE, exactly as for an ordinary Dirichlet condition.

Imposing the value alone is *not* acceptable: the unconstrained Hermite tangent derivative then generates large curvature between nodes, which $w = \Delta^*u$ reads directly.

7 Numerical verification of the normalisation

Evaluated with the constants as `models/constants.f90` defines them ($\mu_0 = 4\pi \times 10^{-7}$, $e = 1.602176565 \times 10^{-19}$, $m_u = 1.660539040 \times 10^{-27}$) at the AUG 38773 parameters `central_density = 1.011088`, `central_mass = 2.01410174369812`, $F_0 = 2.972306$:

quantity	value	
ρ_0	$3.381578331801 \times 10^{-7}$	kg m^{-3}
$\sqrt{\mu_0 \rho_0}$	$6.518754986874 \times 10^{-7}$	
volts per unit u	$4.559622206978 \times 10^6$	V
a_n	$1.856385001490 \times 10^2$	
$C_T = 2\Lambda/a_n$, $\Lambda = 3$	$3.232088168772 \times 10^{-2}$	u per volt
$C_V = \sqrt{\mu_0 \rho_0}/F_0$	$2.193164158359 \times 10^{-7}$	
T_e^{JOREK} for 1 eV	$2.035678524691 \times 10^{-5}$	
u for $T_e = 1$ eV	$6.579492475076 \times 10^{-7}$	
reconstructed $V_p - V_{\text{wall}}$	3.000000000000	V

The round trip closes exactly, and the identity $2e\mu_0 n_0/a_n = \sqrt{\mu_0 \rho_0}/F_0$ holds to all printed digits, which checks the algebra of (9) independently.

Use the code's constants, not CODATA-2018. `constants.f90` carries $e = 1.602176565 \times 10^{-19}$ and $m_u = 1.660539040 \times 10^{-27}$, which are the 2010 values. A unit test written with modern constants disagrees in the 8th significant figure — close enough to look like a rounding difference and far enough to mask a real error.

8 What this derivation does *not* establish

- That the discrete problem is stable. The constraint is exact and its Jacobian is constant, but u and $w = \Delta^*u$ form a mixed system and freezing boundary w at a restart value is not a complete treatment of an evolving u trace.
- That the resulting drift is resolved. $\mathbf{v}_E \cdot \mathbf{n} \propto R C_T \partial_\ell T_e$ is a real normal flow; whether the density advection it drives sits inside the stable cell-Péclet range of the central scheme is a separate, measurable question.
- That a frozen ψ boundary is compatible with that normal drift.

Each is an acceptance test, not an assumption, and each is measured rather than assumed in the implementation plan.